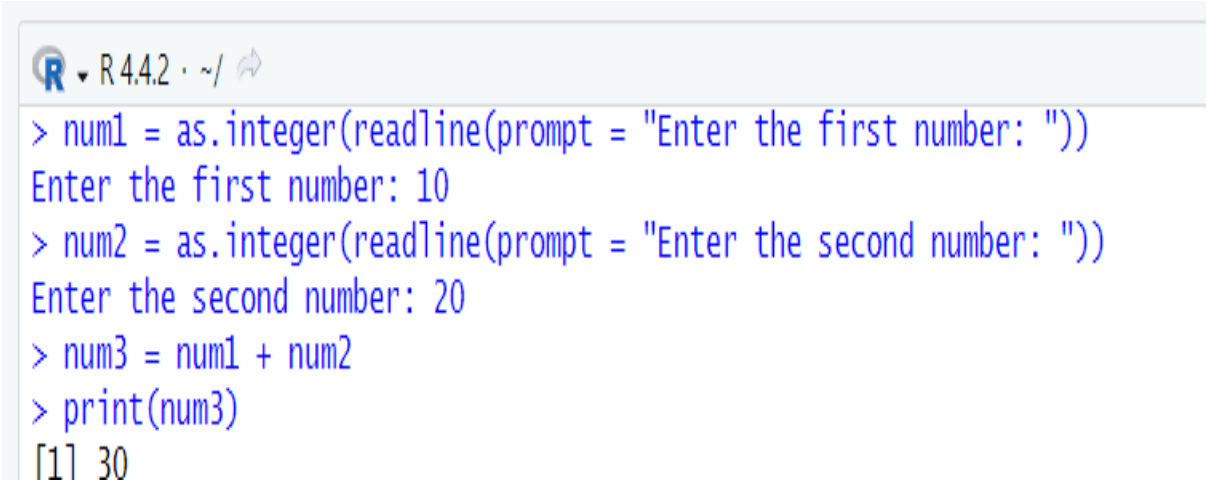
R PROGRAMMING

EXPERIMENT – 1 ADDITION:

AIM:

To prove the program for addition using R-tool.

PROGRAM:

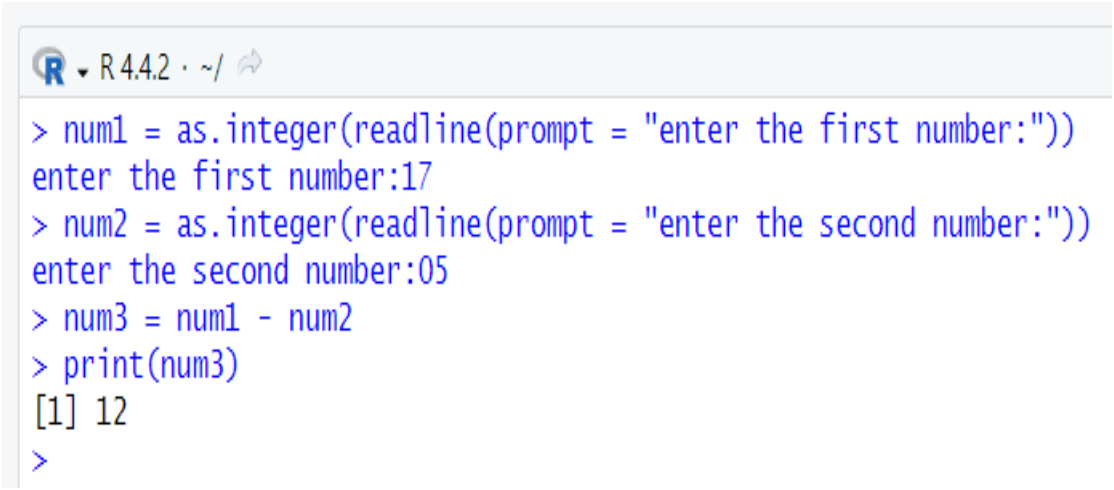


EXPERIMENT – 2 SUBTRACTIONS:

AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

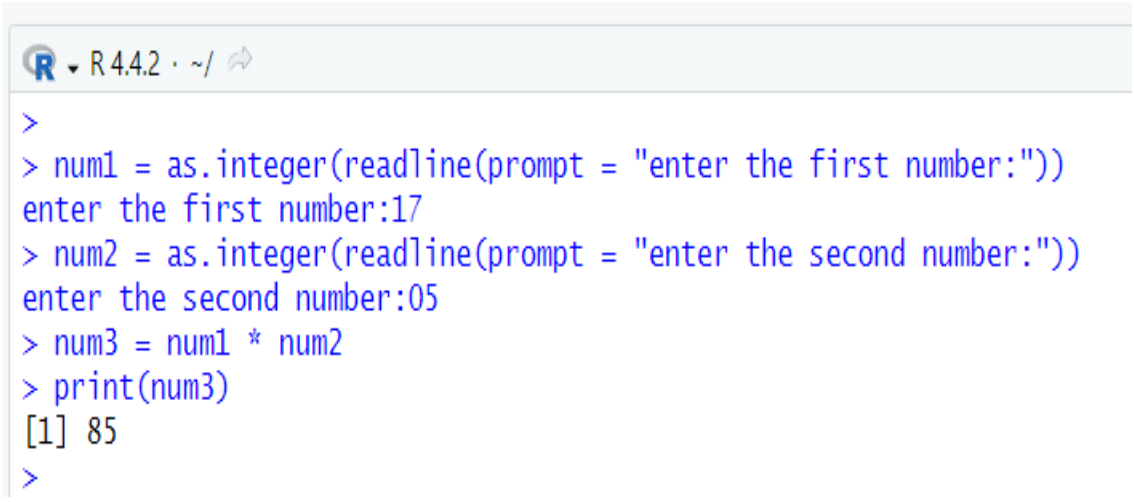


EXPERIMENT – 3 MULTIPLICATION:

AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

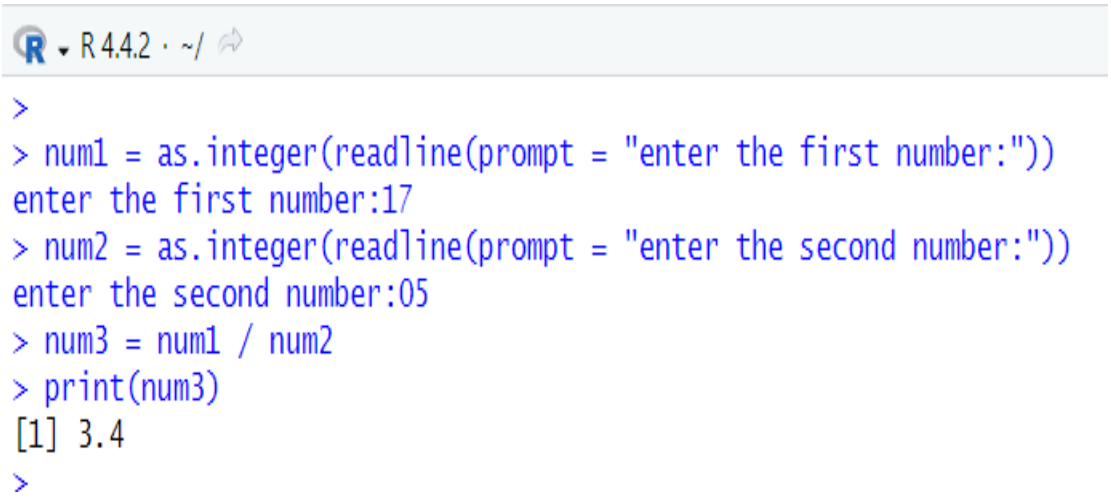


EXPERIMENT – 4 DIVISION:

AIM:

To prove the program for division using R-tool.

PROGRAM:



EXPERIMENT – 5 ODD OR EVEN:

AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> num = as.integer(readline(prompt = "enter a number:"))
enter a number:1756
> if ((num %% 2) == 0)
+ {
+     print("number is even")
+ } else {
+     print("number is odd")
+ }
[1] "number is even"
>
```

EXPERIMENT – 6 MEAN,MEDIAN,MODE:

AIM:

To write the program for mean,median,mode.

PROGRAM:

```
R 4.4.2 · ~/
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mean(df$age)
[1] 24
>
> write.csv(df, "datafr.csv")
>
>
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> median(df$age)
[1] 24
>
> write.csv(df, "datafr.csv")
>
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mode(df$age)
[1] "numeric"
>
> write.csv(df, "datafr.csv")
>
> get_mode <- function(x) {
+     ux <- unique(x)
+     ux[which.max(tabulate(match(x, ux)))]
+ }
>
> get_mode(df$age)
[1] 23
```

EXPERIMENT – 6 SUMMARY:

AIM:

To write the program for summary using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> summary(df$age)
   Min. 1st Qu.  Median    Mean
   23.0    23.5    24.0    24.0
3rd Qu.    Max.
   24.5    25.0
```

EXPERIMENT – 8 GREATER AMONG THREE NUMBERS:

AIM:

To write the program for the greatest among three numbers.

PROGRAM:

```
R 4.4.2 · ~/
> write.csv(df, "datafr.csv")
>
> x <- as.integer(readline(prompt = "Enter first number :"))
Enter first number :17
> y <- as.integer(readline(prompt = "Enter second number :"))
Enter second number :05
> z <- as.integer(readline(prompt = "Enter third number :"))
Enter third number :06
> if (x > y && x > z) {
+     print(paste("Greatest is :", x))
+ } else if (y > z) {
+     print(paste("Greatest is :", y))
+ } else {
+     print(paste("Greatest is :", z))
+ }
[1] "Greatest is : 17"
```

EXPERIMENT – 9 IQR:

AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
R 4.4.2 · ~/
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> IQR(df$age)
[1] 1
>
```

EXPERIMENT – 10 QUANTILE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
R 4.4.2 · ~/
> write.csv(df, "datafr.csv")
>
>
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> quantile(df$age)
  0%  25%  50%  75% 100%
23.0 23.5 24.0 24.5 25.0
>
```

EXPERIMENT – 11 MID RANGE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
R 4.4.2 · ~/
> write.csv(df, "datafr.csv")
>
>
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
```

EXPERIMENT – 12 Z-SCOORE NORMALIZATION:

AIM:

To write the program for Z-scoore normalization using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> A <- c(diabetest1$Age)
>
> Mean <- mean(A, na.rm = TRUE)
> Std <- sd(A, na.rm = TRUE)
>
> Zscore <- (A - Mean) / Std
>
> print(Zscore)
 [1] -1.48875532  0.29612844 -0.43405128  0.86404600 -1.00196884
 [6] -0.02839588  1.51309465 -0.59631344  0.53952168 -1.16423100
[11]  1.10743924 -0.19065804 -0.83970668  1.75648789  0.21499736
[16] -1.56988640  0.70178384 -0.35292020  1.35083248 -0.67744452
```

EXPERIMENT – 13 MIN, MAX, MEAN, MINMAX:

AIM:

To write the program for the minimum, maximum, mean and minmax using r-TOOL

PROGRAM:

```
R 4.4.2 · ~/
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> A <- c(diabetest1$Age)
>
> Mean <- mean(A, na.rm = TRUE)
>
> print(Mean)
[1] 41.35
>
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> A <- c(diabetest1$Age)
>
> Minimum <- min(A, na.rm = TRUE)
>
> print(Minimum)
[1] 22
>
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> A <- c(diabetest1$Age)
>
> Maximum <- max(A, na.rm = TRUE)
>
> print(Maximum)
[1] 63
>
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> A <- c(diabetest1$Age)
>
> Minimum <- min(A, na.rm = TRUE)
> Maximum <- max(A, na.rm = TRUE)
>
> MinMax <- (A - Minimum) / (Maximum - Minimum)
>
> print(MinMax)
 [1] 0.02439024 0.56097561 0.34146341 0.73170732 0.17073171 0.46341463 0.92682927 0.29268293 0.63414634 0.12195122 0.80487805 0.41463415
[13] 0.21951220 1.00000000 0.53658537 0.00000000 0.68292683 0.36585366 0.87804878 0.26829268
```

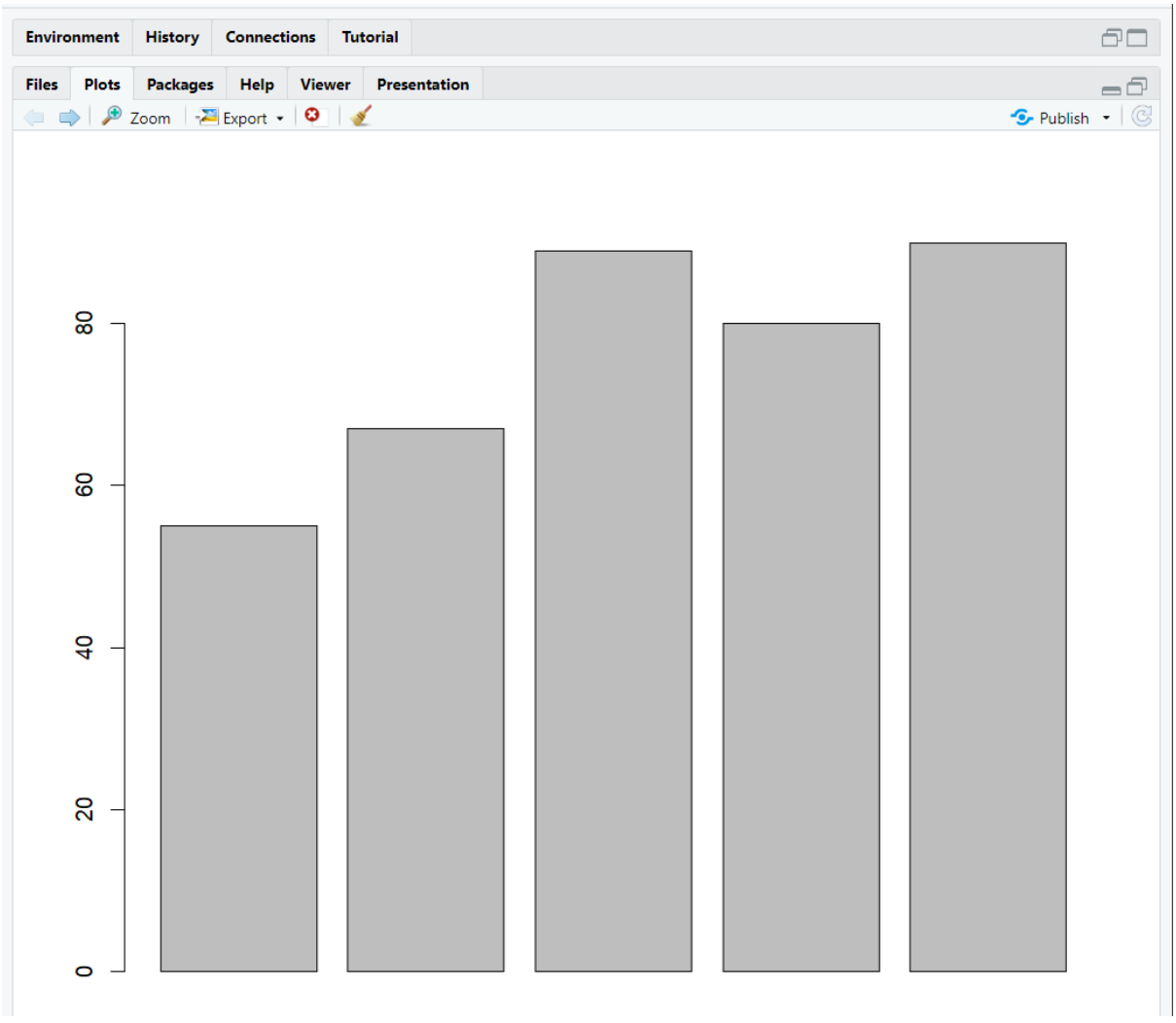
EXPERIMENT – 14 BAR PLOT AND HORIZONTAL BAR:

AIM:

To draw the bar plot and horizontal bar using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> a <- c(55, 67, 89, 80, 90)
> barplot(a)
>
```



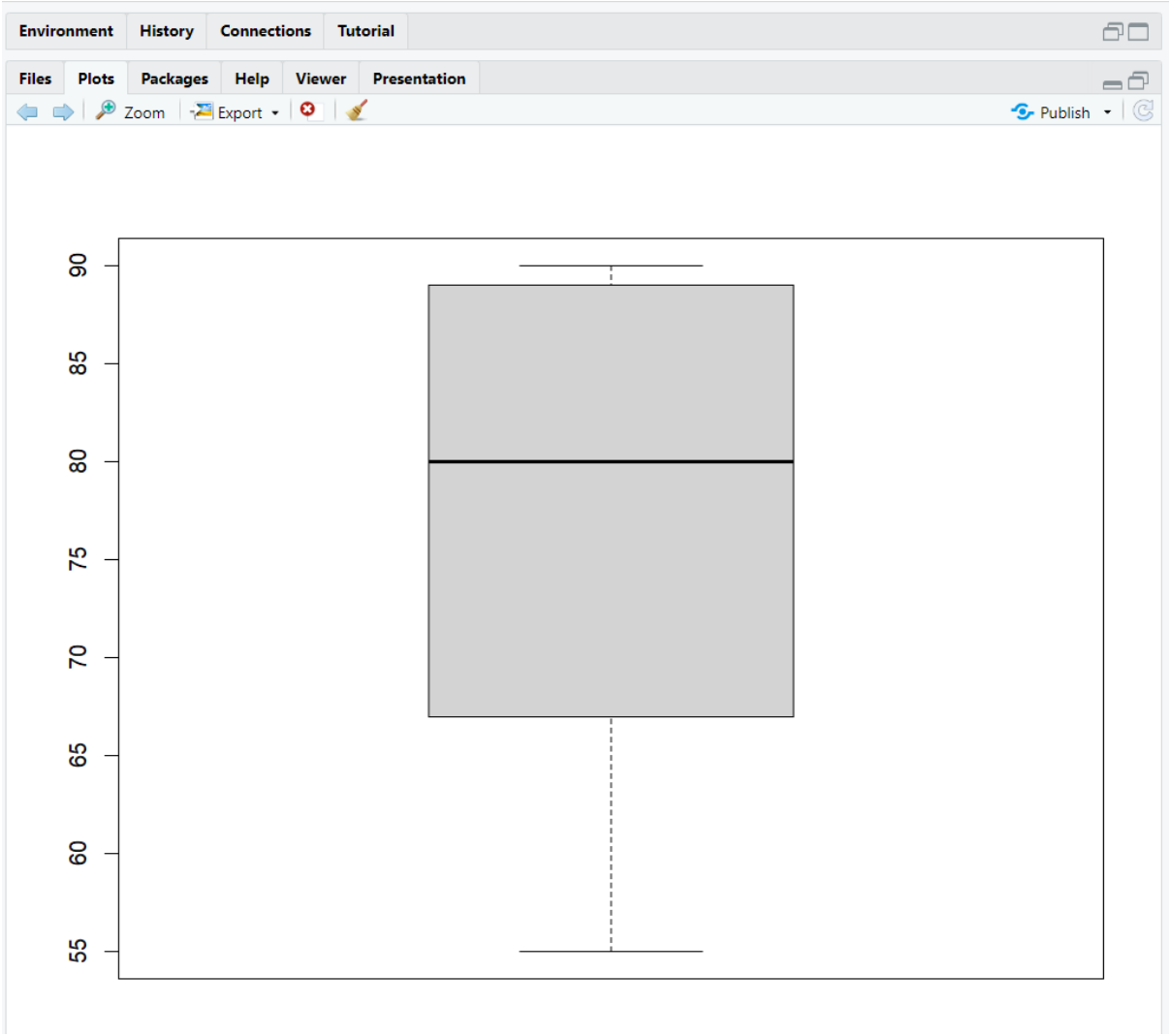
EXPERIMENT – 15 BOX PLOT

AIM:

To draw a box plot using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> a <- c(55, 67, 89, 80, 90)
> boxplot(a)
```



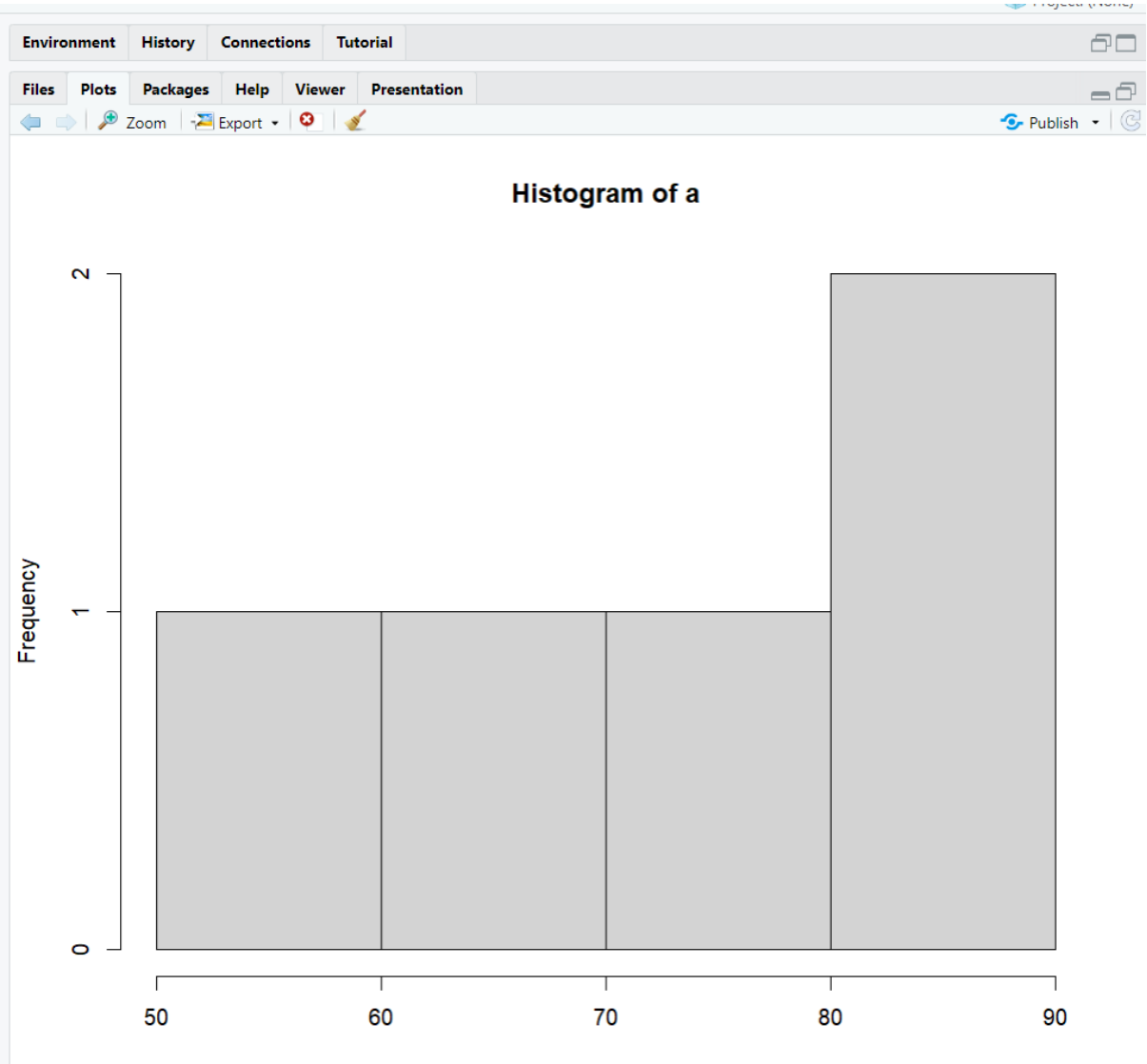
EXPERIMENT – 16 HISTOGRAM

AIM:

To draw a histogram plot using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> a <- c(55, 67, 89, 80, 90)
> hist(a)
>
```



EXPERIMENT – 17 CORRELATION ANALYSIS:

AIM:

To write the program for correlation analysis using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> diabetest1 <- table(
+     diabetest1$Age,
+     diabetest1$DiabetesStatus
+ )
>
> diabetest1

    No Yes
  22  1   0
  23  1   0
  27  1   0
  29  1   0
  31  1   0
  33  1   0
  34  1   0
  36  1   0
  37  1   0
  39  1   0
  41  0   1
  44  0   1
  45  0   1
  48  0   1
  50  0   1
  52  0   1
  55  0   1
  58  0   1
  60  0   1
  63  0   1
>
> chisq.test(diabetest1)

        Pearson's Chi-squared test

data:  diabetest1
X-squared = 20, df = 19, p-value = 0.3946
```

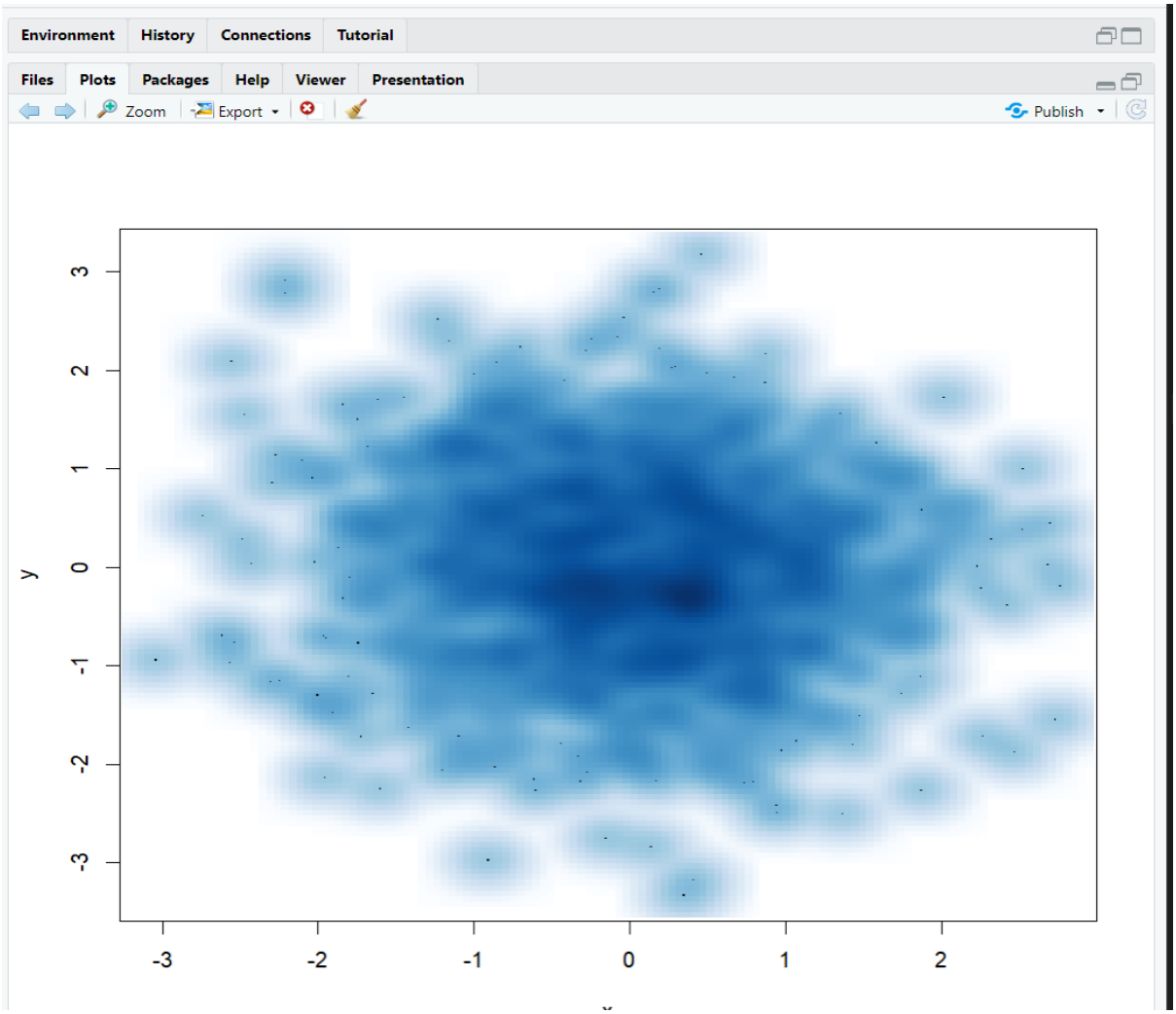
EXPERIMENT – 18 SCATTER PLOT

AIM:

To draw the scatter plot using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> set.seed(9)
>
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> smoothScatter(y - x)
>
> smoothScatter(x, y)
```



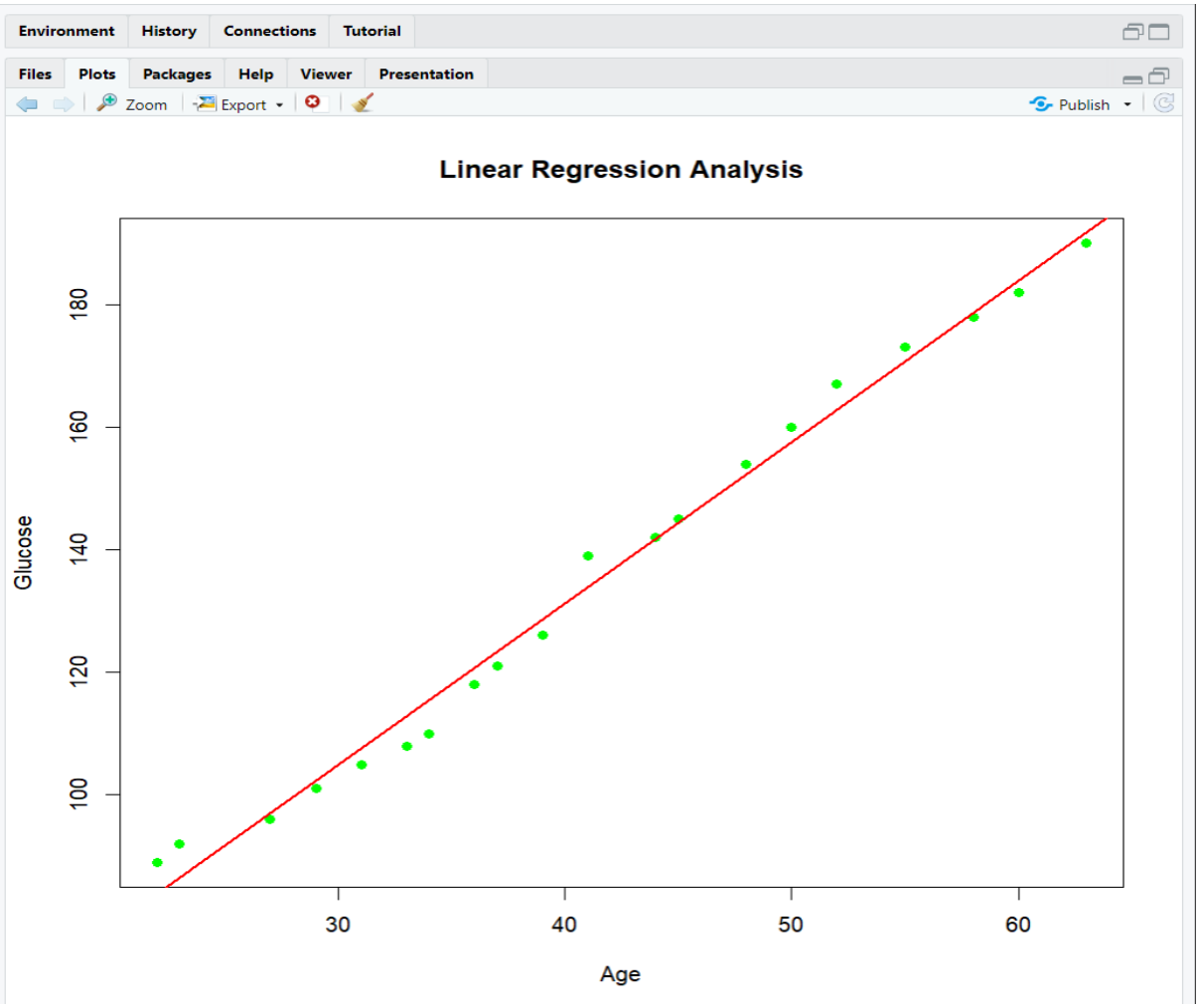
EXPERIMENT – 19 LINEAR REGRESSION

AIM:

To write a program for linear regression using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> library(readxl)
>
> diabetes <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> Relation <- lm(Glucose ~ Age, data = diabetes)
>
> plot(diabetes$Age, diabetes$Glucose,
+      col = "green",
+      main = "Linear Regression Analysis",
+      xlab = "Age",
+      ylab = "Glucose",
+      pch = 19)
>
> abline(Relation, col = "red", lwd = 2)
```



EXPERIMENT – 20 MULTIPLE REGRESSION

AIM:

To write a program for multiple regression using R-tool.

PROGRAM:

```
R 4.4.2 · ~/
> diabetes <- read_excel("C:/Users/rakin/Downloads/NARA.xlsx")
>
> input <- diabetes[, c("Age", "Glucose", "BMI")]
>
> model <- lm(Age ~ Glucose + BMI, data = input)
>
> print(model)

Call:
lm(formula = Age ~ Glucose + BMI, data = input)

Coefficients:
(Intercept)      Glucose          BMI
   -15.4853       0.2811       0.6460

>
> A <- coef(model)[1]
>
> print(A)
(Intercept)
  -15.48529
>
>
> xGlucose <- coef(model)[2]
> yBMI <- coef(model)[3]
>
> print(xGlucose)
  Glucose
0.2811104
> print(yBMI)
      BMI
0.6460306
>
>
> y <- paste(
+     "Age =",
+     round(coef(model)[1], 3), "+",
+     round(coef(model)[2], 3), "* Glucose +",
+     round(coef(model)[3], 3), "* BMI"
+ )
>
> print(y)
[1] "Age = -15.485 + 0.281 * Glucose + 0.646 * BMI"
```